

PLACEMENT BROCHURE

B.TECH – Class of 2027
Computational Engineering,
B.TECH
IIT Hyderabad



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About IIT-H

Indian Institute of Technology Hyderabad (IITH) is a premier institute of science and technology established in 2008. IITH has been consistently ranked among the top 10 engineering institutes in India according to NIRF. IITH offers undergraduate programs in classical engineering disciplines, applied sciences, design, and interdisciplinary areas.

The very foundation of IIT Hyderabad is based on research and innovation. IITH encourages and supports innovation and entrepreneurship at every stage.

IITH enjoys a very special relationship with Japanese universities and industries that goes beyond academic and research collaborations. IITH creates a holistic ecosystem for education that enables students and faculty to translate their dreams into realities.

Vision

IITH will be the cradle for inventions and innovations. It will advance knowledge and scholarship to students in science, technology and liberal arts, and equip them to handle the challenges of the nation and the world in the 21st century.

Mission

IITH will be the cradle for inventions and innovations. It will advance knowledge and scholarship to students in science, technology and liberal arts, and equip them to handle the challenges of the nation and the world in the 21st century.



ABOUT COMPUTATIONAL ENGINEERING

Launched in 2021, this is India's first undergraduate program tailored to meet the computational demands of modern industry.

Why Computational Engineering at IITH?

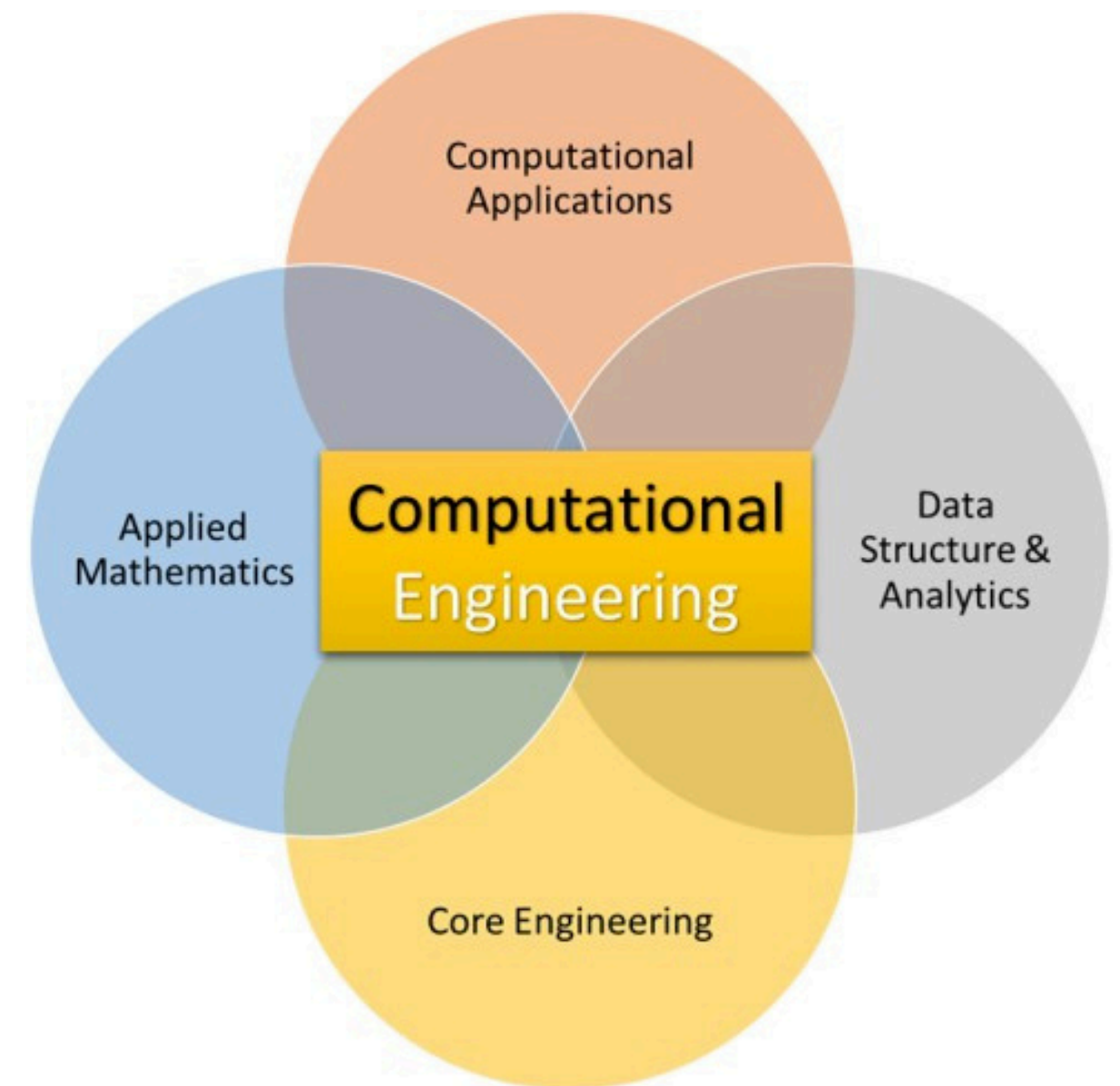
- Bridges traditional engineering with advanced computing, AI, and automation.
- Aligned with Industry 4.0 and digital transformation needs.
- Designed to build engineers fluent in both core engineering and computational tools.

Program Highlights

- Interdisciplinary Curriculum based on four pillars:
 - Applied Mathematics
 - Core Engineering
 - Data Structures & Analytics
 - Computational Applications
- Key Skills Developed:
 - Scientific computing, machine learning, optimization
 - Numerical methods, simulations, hardware description languages
- T-shaped Learning: Strong foundational breadth with options for computational specialization.

Career Readiness

- Focus on real-world problem solving, digital twin creation, and systems-level modeling.
- Prepares graduates for roles in:
 - Aerospace, energy, manufacturing
 - Healthcare, finance, and data science
- Ideal for leadership in research, academia, and industry.



MESSAGE FROM OUR HEAD OF DEPARTMENT



Prof. Sathya Peri
Head of Department, Department

Dear Recruiters,
I am honored to invite you to the Computational Engineering program at IIT Hyderabad.

Launched in 2021, this pioneering undergraduate program is designed to equip students with strong fundamentals in engineering alongside cutting-edge computational skills. Through a carefully structured curriculum and an emphasis on hands-on problem solving, our students are well-prepared to address modern industrial challenges.

Our graduates have secured competitive internships at leading organizations such as Google, Adobe, Siemens, and D.E. Shaw, contributing to projects spanning ML-based automation, biomedical systems, operating systems, and simulation technologies.

We foster a culture of creativity, collaboration, and critical thinking. Our students are not only technically sound but also research-oriented and ready to make meaningful contributions from day one.

I extend an invitation to your prestigious organization to partner with our program during this placement period. I believe that our graduates will bring immense value to your teams in the form of their technical expertise, flexibility, and innovative mindset. We eagerly await the chance to interact with you and appreciate your ongoing support.



CURRICULUM

Computer Science & Programming

- Introduction to Programming
- Data Structures & Applications
- Operating Systems
- Parallel & Concurrent Computing
- Distributed Computing
- Design & Analysis of Algorithms
- Formal methods for program verification

Machine Learning & Computational Science

- Foundations of Machine Learning
- Deep Learning
- Fundamentals of Scientific Computing
- Computer-Aided Numerical Methods
- Optimization Techniques
- Biostatistics
- Computational Methods in Material Science

Mathematics and Electrical

- Calculus
- Linear Algebra
- Differential Equations
- Probability
- Transform Techniques
- Complex Variables
- Statistics
- Digital Circuits
- Intro to Hardware Description Language

Core Engineering

- Solid and Fluid Mechanics
- Thermodynamics and Kinetics
- Heat and Mass Transfer
- Computational Fluid dynamics and Finite element method
- Basic Bioinformatics
- Big Data Biology and Biological Databases
- Electronic Structure and Atomistic Modeling

CORE SKILLS

Programming & Software Development

Programming Languages

- C, C++, Python, Java, JavaScript, Kotlin, Julia

Core CS

- Data Structures & Algorithms
- Operating Systems
- Low-Latency Computing in C

Web Development

- Angular
- HTML/CSS

Development Tools

- Git/GitHub
- Collaborative Software Development



Machine Learning, AI & Data Science

Deep Learning Frameworks

- TensorFlow
- PyTorch
- Custom Autograd Frameworks

Artificial Intelligence

- Machine Learning, Deep Learning
- Natural Language Processing (NLP)
- Computer Vision

Data Science & Analytics

- Python, R
- SQL/MySQL
- Data Analysis & Visualization

Generative AI

- LLMs/RAG systems
- AI Application Development



High-Performance & Parallel Computing

Parallel & Distributed Computing

- OpenMP, Pthreads
- Message Passing Paradigms
- Distributed Systems Design

Scientific & Numerical Computing

- MATLAB, Numerical Methods
- Large-Scale Computational Modeling

Cloud & Infrastructure

- Microsoft Azure
- Efficient data exchange via Protocol Buffers

Large-Scale Engineering Applications

- Distributed LU Factorization
- Barnes-Hut N-Body Simulations
- Parallel FEM Solvers



Computational Science, Simulation & Hardware

Scientific computing, numerical methods and simulation across

- Computational Fluid Dynamics (CFD)
- Finite Element Method (FEM)
- Molecular Dynamics (MD)
- Phase-Field Modeling

Design & Engineering Tools

- SolidWorks & CAD

Hardware & Digital Systems

- Verilog, VHDL, HDLs
- Digital System Design

Mathematical Foundations

- Linear Algebra
- Optimization Techniques
- Probability & Statistics



WHY RECRUIT US?

Computational Engineering at IIT Hyderabad combines strong foundations in engineering, mathematics, systems, and AI. Our students are trained to solve large-scale computational problems and contribute across software engineering, machine learning, quantitative systems, and research-oriented roles.



Placement Rate



GATE Mathematics
from Batch'2027



Selective Batch
Size

Students have consistently demonstrated excellence through internships, research projects, competitive programming, and industry collaborations. Their interdisciplinary training enables them to adapt quickly and contribute effectively across diverse technical domains.

Selective and High-Impact Batch: With a focused batch size of **25 students**, recruiters benefit from access to a highly selective and technically versatile talent pool.

Below are the recent cutoff ranks and batch size details:

Year	Opening Rank	Closing Rank	Batch Strength
2021	1171	1383	10
2022	1212	1633	15
2023	1304	1884	25 (Current Placement Batch)
2024	1046	1790	25
2025	1368	1675	24

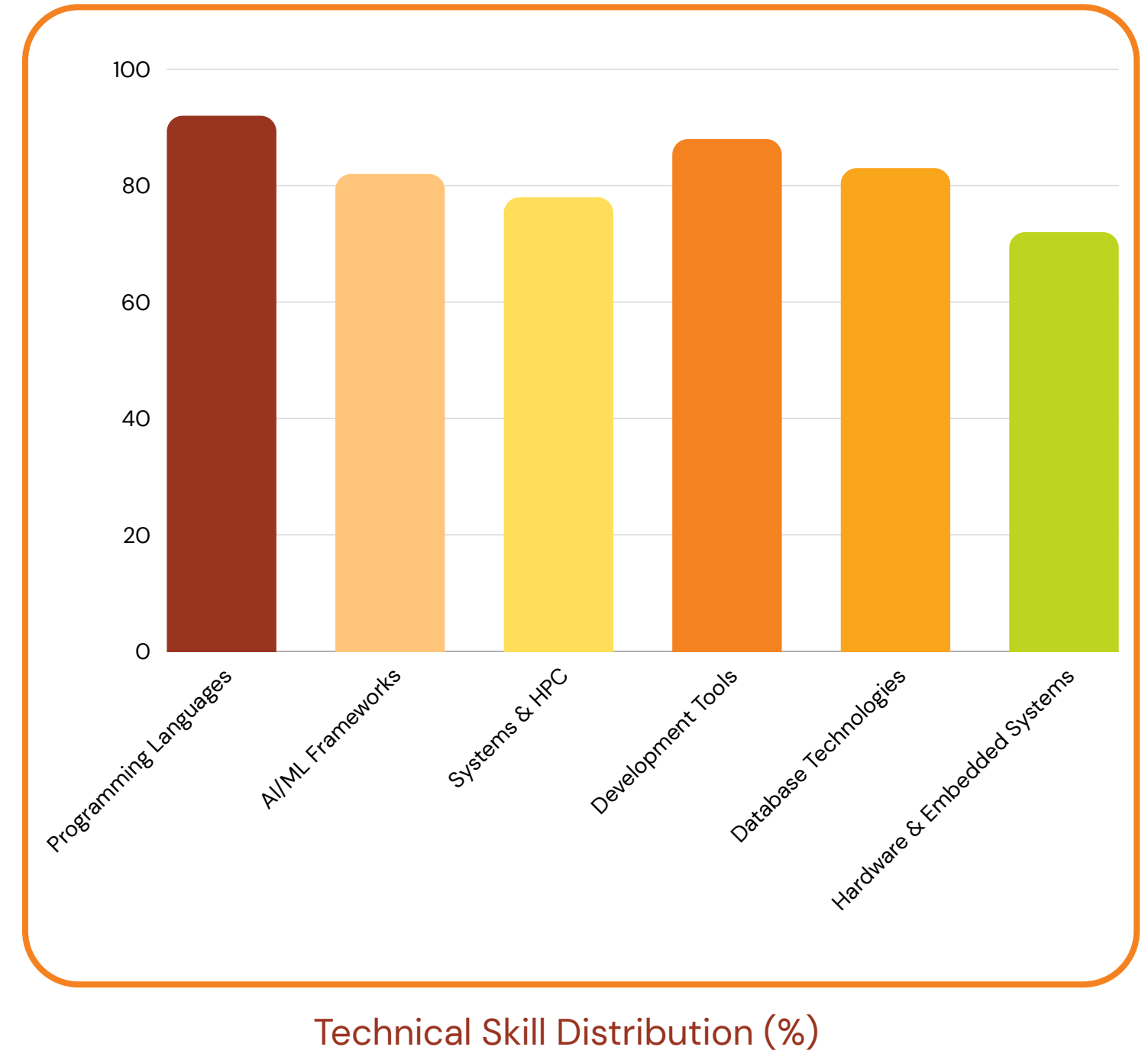
BEYOND THE CURRICULUM

Computational Engineering students apply classroom knowledge to solve real-world challenges through simulations, AI applications, systems programming, computational modeling, and interdisciplinary research.

Why COE Students Stand Out

- ✓ Strong Systems + AI Foundation
- ✓ Expertise in HPC, Parallel Computing, and Simulations
- ✓ Industry-Oriented Project and Research Experience
- ✓ Cross-Domain Readiness for **SWE, ML, Quant, and Systems Roles**
- ✓ Rigorous Mathematical and Engineering Training
- ✓ Strong Project-Based and Real-World Engineering Exposure
- ✓ Proven Internship and PPO Track Record

“Students have secured internships and research opportunities at leading global organizations, demonstrating their ability to contribute to industry-scale challenges from an early stage.”



PROJECTS

A collection of industry-oriented and high-performance computing projects demonstrating expertise in AI/ML, software engineering, optimization, distributed systems, and parallel computing. These projects showcase practical problem-solving, scalable system design, and the application of advanced computational techniques to real-world challenges.

Real-World Internship Projects

- **LLMOps Platform with Agent Automation:** Built a platform integrating ModelOps, AgentOps, LiteLLM, multi-LLM access, virtual API key provisioning, governance, and automated workflow management
- **Optimization-Based Interview Scheduling System:** Designed an interview scheduling system using Google OR-Tools CP-SAT Solver to maximize student interview assignments under complex constraints
- **RAG Model for Software Testing:** Unit testing automation using LLMs is actively explored in the industry
- **Jira Issue Analytics with LLM:** Real-world use case in software project management and trend analysis
- **Android UI Test Automation with CI/CD:** Built and automated regression tests for sort and filter features in Android applications with scheduled CI/CD pipeline execution
- **iOS Regression Automation Framework:** Automated regression testing pipelines for feature validation in iOS applications
- **Fine-Tuned T5 Model for API Endpoints:** Useful for automating developer workflows in backend teams
- **Material Detection using ML:** Industrial vision and automation systems rely on this kind of ML

Parallel and High-Performance Computing

- **Distributed LU Factorization:** Implemented distributed LU decomposition using TCP sockets and pthreads with custom message-passing and block-row parallelization for high-performance computing
- **Parallel N-Body Simulation (Barnes-Hut):** Implemented a parallel algorithm to optimize gravitational simulations using spatial decomposition
- **Parallelization of Finite Element Method (FEM):** Optimized global stiffness matrix assembly using Pthreads and OpenMP
- **Delta-Stepping for the Shortest Path:** Parallelized single-source shortest path computation for large graphs using OpenMP
- **Multithreaded Processing using Pthreads:** Implemented a multithreaded C application using pthreads, mutex synchronization, and concurrent execution for parallel performance optimization

PROJECTS

Driven by innovation and interdisciplinary collaboration, our department undertakes projects in simulation, machine learning, deep learning, data-driven modeling & Simulation, and systems engineering. These efforts contribute to advancing knowledge while developing solutions for real-world challenges.

Computational Science & Simulation

- **GAN-Based Rollout Acceleration for Dendritic Growth:** Used Generative Adversarial Networks to accelerate mesoscale phase-field dendritic growth simulations
- **Dendritic Growth via Phase-Field:** Simulated dendritic solidification with differential analysis of interface points
- **Benchmarking ML Force Fields:** Benchmarked graph neural networks and equivariant message-passing models for material property and atomic force prediction
- **Epidemic Spread Modeling:** Developed a scalable and parallel model using mathematical and computational modeling

Machine Learning & Deep Learning

- **2D to 3D CAD using Transformers:** Generated CAD models from orthographic views using SVG and CAD command sequences
- **Custom ML/DL Framework with Autograd Engine:** Developed a ML and DL library from scratch with a PyTorch-inspired autograd engine, classical ML algorithms, and DL models including MLPs, CNNs, and RNNs
- **Small Language Models for NLP Tasks:** Built compact language models for efficient natural language processing and inference
- **Sensor-Based Human Activity Classification:** Achieved 98% accuracy using CNN-LSTM models for activity recognition from sensor data
- **Audio Classification using RNN, CNN, Transformers:** Built deep learning models to classify audio inputs with high accuracy

Systems & Applications

- **Scalable Ticket Booking System using Microservices:** Built a distributed ticket booking platform using scalable microservice architecture
- **System Call Implementation and Demand Paging in xv6:** Added custom system calls and copy-on-write demand paging features to the xv6 kernel
- **Speech-to-Text Web Application:** Built a web app for real-time speech-to-text conversion using ML techniques.

PREVIOUS STATISTICS

14

STUDENTS
REGISTERED

14

TOTAL STUDENTS
PLACED

100 %

PLACEMENT
RATE

₹65.2
LPA

HIGHEST
PACKAGE

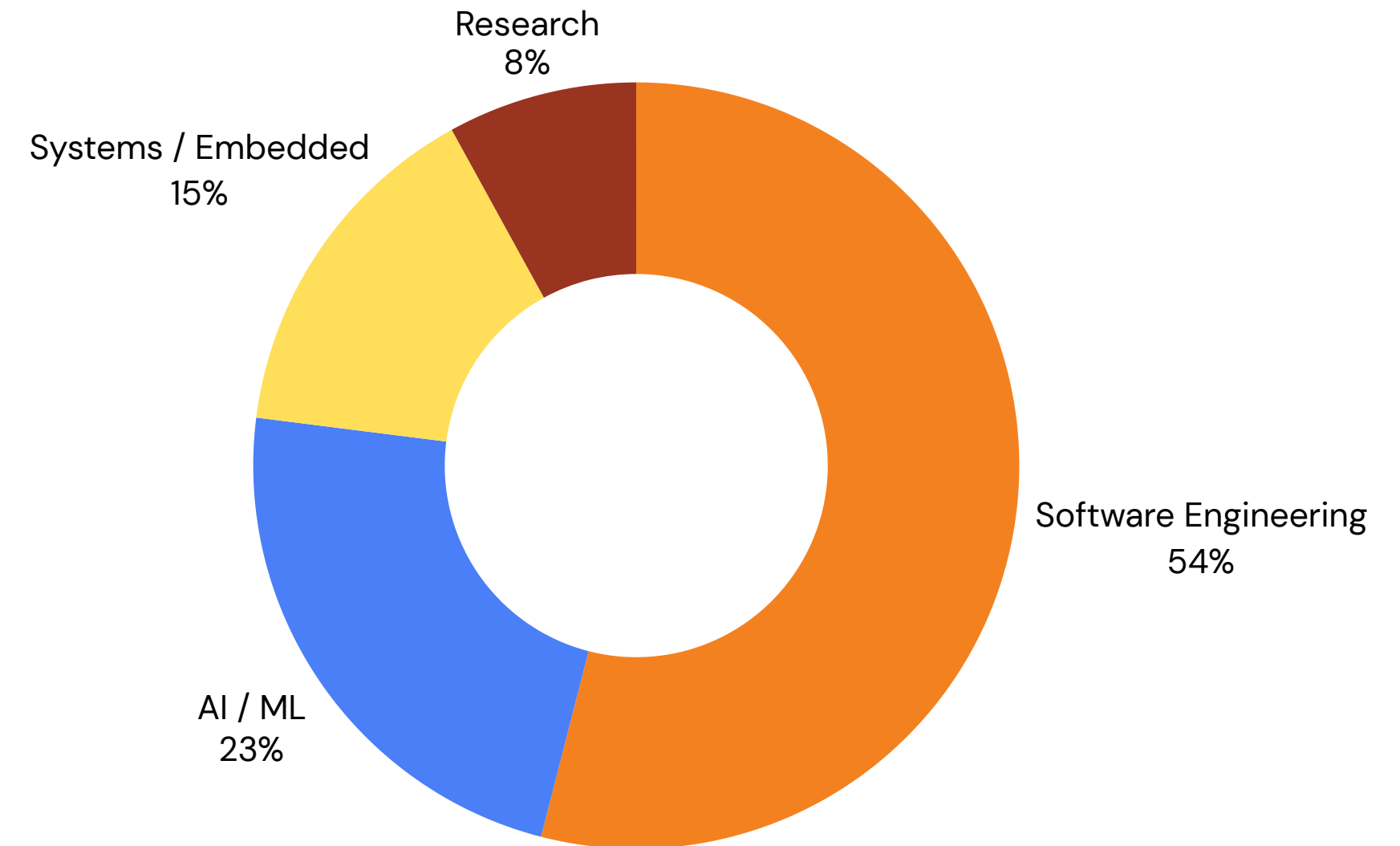
₹2.66
L

HIGHEST
STIPEND '27
BATCH

10+

RECRUITERS

PLACED DOMAINS



PAST RECRUITERS

AMERICAN
EXPRESS

Google

accenture

KLA Tencor



Thomson
Reuters™



Adobe

amazon



NIKKO



ATLASSIAN



Sagri

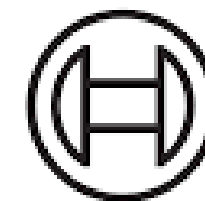
Accelquant



Google Summer of Code

DE Shaw & Co

SIEMENS



BOSCH

PAST RECRUITERS



teradata.



Jupiter

Meltwater



muratec



CHUBB®

VERINT®

Deutsche Bank



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CONTACT US

OFFICE OF CAREER SERVICES (OCS)

The Office of Career Services (OCS) at IIT Hyderabad is committed to empowering students to make informed and strategic career decisions aligned with their strengths and aspirations. By fostering strong collaborations with industry leaders and leveraging a robust alumni network, OCS facilitates a broad spectrum of career opportunities.

Through well-structured guidance initiatives and a streamlined recruitment process, OCS effectively bridges the gap between academic preparation and professional expectations, ensuring that students are equipped to excel in roles that best suit their career goals.



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